5.2 Solving Systems of Linear Equations by Substitution

Name _____ Period ___ Date _____

Algebra I 5.2 · Student sheet

I can…

- solve a system of linear equations by substitution.
- choose which variable to isolate to make substitution easier.
- solve real-life problems using substitution.

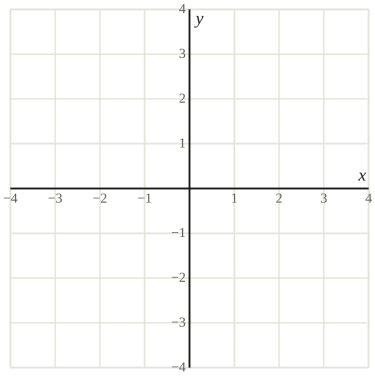
Vocabulary · fill in the missing word

Substitution — solve one _____ for a variable, then replace that variable in the other equation

Part 1 · Graph vs. substitute

y = 3x − 1 y = −x + 2.2

Solve by graphing:



Solve by substitution:

Which method was more exact? Why? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Step 1: Step 1: solve one equation for one _____.

Step 2: Step 2: _____ that expression into the _____ equation.

Example 1

Solve.

y = 3x − 1

2x + y = 9

Careful: Use _____ when you substitute.

Example 2

Solve.

$x - 2y = 1$

$3x + 4y = 23$

Example 3

A theater sells 150 tickets. Adult tickets cost $10 and child tickets cost $6. The total is $1180. How many of each were sold?

Part 3 · Partner practice

Partner 1

Solve: $y = x + 4$ and $3x - y = 6$.

Partner 2

Solve: $2x + y = 7$ and $x - 3y = 7$.